

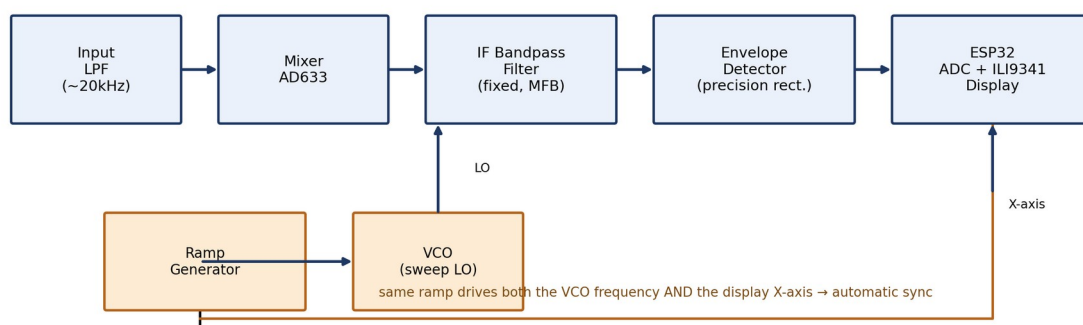
Analog Heterodyne Spectrum Analyzer

IF Filter, Mixer, VCO & Sweep Design Reference — with worked values

This document works through every block between the mixer and the display, with real component values derived from verified design equations, and full datasheet/reference links. Values marked "starting point" should be confirmed in LTspice before breadboarding — component tolerances and real op-amp behavior will need small trims.

1. System Recap

Analog Heterodyne Spectrum Analyzer — Signal Chain



Chosen design values

Parameter	Value	Reasoning
Input range	20 Hz – 20 kHz	Standard audio spectrum
IF frequency (f_{IF})	25 kHz	Above input range → up-conversion gives image rejection without a preselector
LO sweep range	25.02 kHz – 45 kHz	$f_{LO} = f_{in} + f_{IF}$ across the input range
Filter Q	15	Balances resolution against buildability with a single MFB stage
Resolution bandwidth (BW)	≈ 1.67 kHz	$BW = f_{IF} / Q$
Sweep time (T_{sweep})	200 ms (5 Hz refresh)	$\approx 10\times$ the calculated minimum — see Section 8

2. Block 1 — Input Low-Pass Filter

Simple single-pole RC low-pass, cutoff just above the input band. Purpose: keep anything above 20kHz from ever reaching the mixer (cheap insurance alongside the up-conversion IF choice).

$$f_c = 1 / (2\pi R C)$$

Component	Value	Notes
R	10 k Ω	Standard value
C	820 pF	Gives $f_c \approx 19.4$ kHz (closest standard value to the 20kHz target)

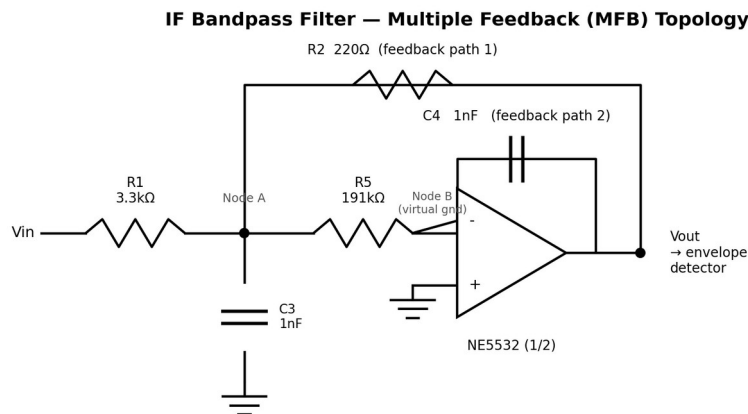
3. Block 2 — Mixer (AD633)

The AD633 computes $W = (X1-X2)(Y1-Y2)/10V + Z$. Connect your filtered input signal to X1 (X2 to ground), your VCO output to Y1 (Y2 to ground), and Z to ground (no offset needed). W is your mixer output, carrying both sum and difference frequencies.

Pin	Connect to	Notes
X1	Input signal (from LPF)	Keep signal amplitude in the volt range, not mV, for good SNR
X2	Ground	—
Y1	VCO output	LO drive — also volt-range for good conversion gain
Y2	Ground	—
Z	Ground	No DC offset needed
W	→ IF Bandpass Filter input	Output — contains $f_{LO}-f_{in}$ and $f_{LO}+f_{in}$

4. Block 3 — IF Bandpass Filter (MFB Topology)

Multiple Feedback (MFB) bandpass — the standard single-op-amp choice for Q below ~20 (Analog Devices MT-218, Section 8). Two feedback paths give the topology its name: R2 (resistive, node A → output) and C4 (capacitive, node B → output).



$f_0 = 25 \text{ kHz}$ | $Q = 15$ | $BW \approx 1.67 \text{ kHz}$ | Mid-band gain $H = 2$ (6 dB)
Two feedback paths (R2 and C4) — hence “multiple feedback”. Use NE5532 (not TL072) — sufficient GBW margin at this Q.

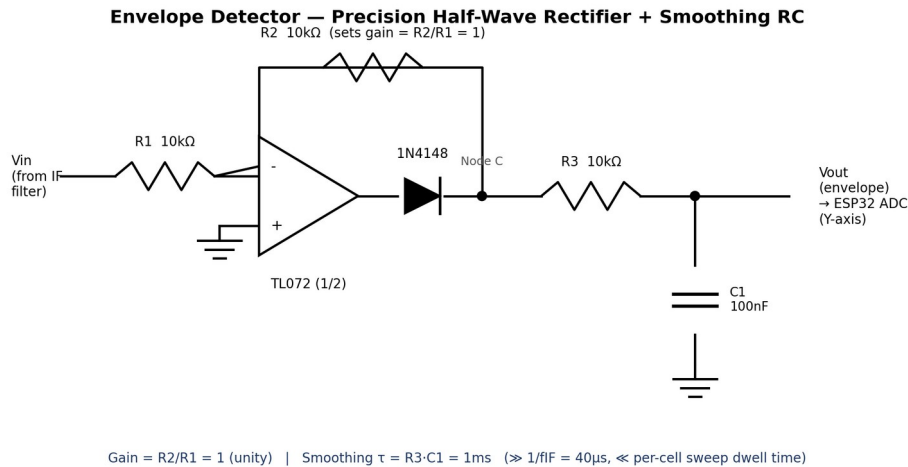
$$k = 2\pi f_0 C \quad R1 = 1/(H \cdot k) \quad R2 = 1/(k \cdot (2Q - H)) \quad R5 = 2Q/k$$

Component	Value	Formula result	Notes
C3, C4	1 nF	chosen first	Both capacitors equal — standard simplification
R1	3.3 kΩ	3183 Ω calculated	Sets gain path
R2	220 Ω	227 Ω calculated	Feedback path 1
R5	190 kΩ	190,986 Ω calculated	Feedback path 2 — make this a trim pot to fine-tune f_0
Op-amp	NE5532	—	TL072's GBW is marginal at $Q=15$, $f_0=25\text{kHz}$ — see check below

⚠ Op-amp gain-bandwidth check: peaking $\approx 20 \cdot \log_{10}(Q) = 23.5\text{dB}$, plus a 20dB safety margin (per AD MT-218) means the op-amp needs $\geq 43.5\text{dB}$ open-loop gain at 25kHz. TL072 (3MHz GBW) only gives $\approx 41.6\text{dB}$ there — marginal. NE5532 (10MHz GBW) gives $\approx 52\text{dB}$ — comfortable margin. Use NE5532 for this stage specifically.

5. Block 4 — Envelope Detector (Precision Rectifier + Smoothing)

A bare diode fails to rectify small signals cleanly (its $\sim 0.6\text{V}$ forward drop dominates). Putting the diode inside an op-amp feedback loop removes that error. The RC stage afterward smooths the rectified ripple into a slowly-varying envelope.

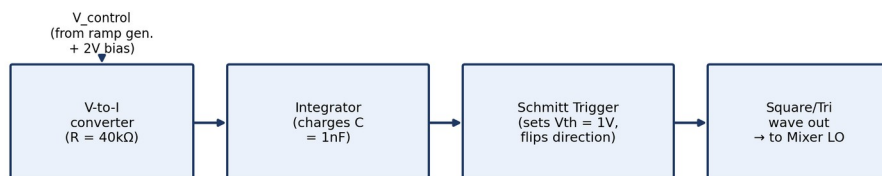


Component	Value	Notes
R1, R2	10 kΩ each	Gain = $R_2/R_1 = 1$ (unity) — change ratio if more gain is needed
Diode	1N4148	Fast small-signal diode, well suited to this frequency range
R3, C1 (smoothing)	10 kΩ, 100 nF	$\tau = R_3 \cdot C_1 = 1\text{ms}$ — smooths the 25kHz IF ripple (period $40\mu\text{s}$) while staying much faster than the sweep dwell time per resolution cell
Op-amp	TL072	GBW is not the bottleneck here — this stage is fine with TL072

6. Block 5 — VCO (Sweeping Local Oscillator)

A relaxation oscillator: integrator + Schmitt trigger. The Schmitt trigger flips the integrator's charge direction each time the capacitor voltage hits a threshold, producing a triangle/square wave whose frequency is proportional to the control current (and therefore control voltage).

Sweeping Local Oscillator (VCO) — Functional Block Diagram



$$f_{\text{VCO}} = V_{\text{control}} / (2 \cdot R \cdot C \cdot V_{\text{th}}) \quad \rightarrow \text{linear: } V_{\text{control}} = 2\text{V gives } 25\text{kHz}, V_{\text{control}} = 3.6\text{V gives } 45\text{kHz}$$

$R = 40\text{k}\Omega$, $C = 1\text{nF}$, $V_{\text{th}} = 1\text{V}$ — starting values: simulate in LTspice and trim R with a potentiometer to hit exact range

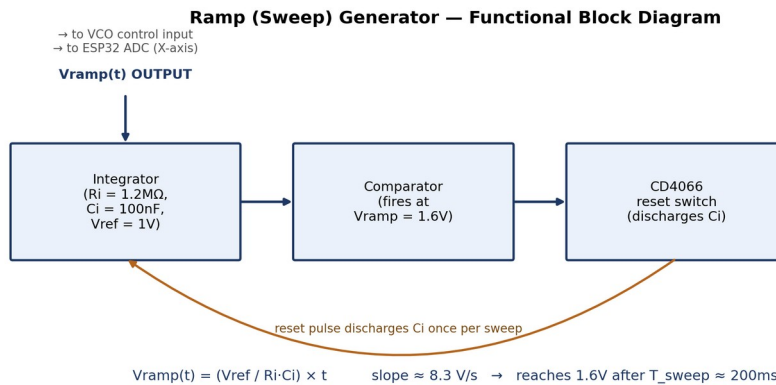
$$f_{VCO} = V_{control} / (2 \cdot R \cdot C \cdot V_{th})$$

Component	Value	Notes
R (V-to-I)	40 kΩ	Use ~39kΩ standard, or a trimmer for exact calibration
C (integrator)	1 nF	—
V _{th}	1 V	Set by the Schmitt trigger's threshold-setting resistor divider
V _{control} range	2 V – 3.6 V	2V = fixed bias (gives 25kHz alone), plus 0–1.6V from the ramp generator on top

Good exercise: verify this linear relationship in LTspice by sweeping V_{control} from 2V to 3.6V and confirming the output frequency tracks 25kHz→45kHz linearly, not just at the two endpoints.

7. Block 6 — Ramp Generator (Sweep + Display X-axis)

A free-running integrator with periodic reset. The same output drives both the VCO's control input and the ESP32's X-axis ADC channel — since both derive from the same ramp voltage, they stay synchronized automatically.



$$V_{ramp}(t) = (V_{ref} / (R_i \cdot C_i)) \times t$$

Component	Value	Notes
R _i	1.2 MΩ	Large value keeps the charging current small and low-power
C _i	100 nF	—
V _{ref}	1 V	Reference feeding the integrator — a simple resistor-divider reference is enough
Reset switch	CD4066	Bilateral analog switch — briefly shorts C _i once V _{ramp} hits 1.6V, triggered by a comparator
Comparator (reset trigger)	TL072 or LM311	Compares V _{ramp} against a 1.6V reference, fires the CD4066 reset

8. Sweep Time Calculation

Real spectrum analyzers relate sweep time, span, and resolution bandwidth with a well-known formula — narrower resolution (better selectivity) forces a slower sweep, since a narrow filter needs longer to settle at each frequency point.

$$T_{sweep} \geq k \times \text{Span} / \text{RBW}^2$$

Where k is a settling-time safety factor, typically 2–4 for a single-pole analog RBW filter (Rohde & Schwarz, Section 9).

Quantity	Value	Source
Span	20,000 Hz	20Hz–20kHz input range
RBW (= BW)	1,666.7 Hz	From Section 1: f_{IF} / Q
k	3	Conservative factor for a single-stage MFB filter
T_{sweep} (minimum)	≈ 21.6 ms	Calculated: $3 \times 20000 / 1666.7^2$
T_{sweep} (recommended)	200 ms	$\sim 10\times$ margin — gives a visible 5Hz trace refresh rate, well clear of the minimum

At $T_{\text{sweep}} = 200\text{ms}$, your ESP32 needs to sample fast enough to fill the display width smoothly: $200\text{ms} / 240 \text{ pixels} \approx 833\mu\text{s}$ per pixel, i.e. ADC sampling at $\geq 1.2\text{kHz}$ — trivial for the ESP32's ADC, even in simple (non-DMA) single-read mode.

9. Consolidated Bill of Materials

Component	Qty	Value / Part
AD633 analog multiplier	1	Mixer
NE5532 dual op-amp	1	IF bandpass filter (GBW-critical stage)
TL072 dual op-amp	2–3	Envelope detector, VCO integrator/comparator, ramp generator/comparator
1N4148 signal diode	1+	Precision rectifier
CD4066 quad bilateral switch	1	Ramp reset
Resistors	assorted	10k Ω , 3.3k Ω , 220 Ω , 190k Ω (or trim pot), 40k Ω , 1.2M Ω , plus standard gain-setting resistors
Capacitors	assorted	820pF, 1nF $\times 3$, 100nF $\times 2$
ESP32 dev board	1	ADC sampling + display driver
ILI9341 240 $\times$ 320 SPI TFT	1	Display

10. Links & Datasheets

Mixer

- [AD633 product page & datasheet — Analog Devices](#) — Official

Op-amps

- [TL072 datasheet — Texas Instruments \(official\)](#)
- [NE5532 — search TI.com product page for current datasheet](#) — Higher GBW than TL072 — needed for the IF filter stage

Diode & switch

- [1N4148 — onsemi official product/datasheet page](#)
- [CD4066B datasheet — Texas Instruments \(official\)](#)

Filter design theory

- [MT-218: Multiple Feedback Band-Pass Design Example — Analog Devices \(official\)](#) — Source of the exact R1/R2/R5 design equations used in Section 4
- [SLOA049: Active Filter Design — Texas Instruments](#) — Broader filter design reference (Sallen-Key and MFB)
- [Multiple Feedback Bandpass Filter — sound-au.com](#) — Practical walkthrough with a spreadsheet-style design approach

Sweep time / resolution bandwidth

- [Understanding basic spectrum analyzer operation — Rohde & Schwarz](#) — Explains why RBW forces the sweep-time trade-off
- [AN 1318: Optimizing Spectrum Analyzer Measurement Speed — Agilent/Keysight](#) — Deeper dive, includes the k-factor discussion

ESP32 / display

- [ESP32 ADC — official Espressif ESP-IDF documentation](#)
- [ESP32 ADC Continuous \(DMA\) mode driver — official docs](#) — Useful if single-read ADC isn't fast enough

Simulation & fabrication

- [LTspice — Analog Devices \(free download\)](#) — Simulate every block in this document before breadboarding
- [JLCPCB](#) — PCB fabrication

11. Notes Before You Simulate

- Every R/C value above is a starting point derived from ideal-op-amp formulas — real components have tolerance (typically $\pm 5\%$), so expect to trim R5 (filter tuning) and R (VCO) once on the bench.
- Simulate each block alone first (LPF, then mixer, then filter, then envelope detector, then VCO, then ramp generator) before connecting them — isolates bugs to one stage at a time.
- The GBW check in Section 4 is the kind of calculation worth including in your project report — it shows you understood a real design constraint, not just copied values.
- Confirm your supply rails (dual $\pm 12\text{V}$ vs single-supply with a virtual ground) early — several of these circuits assume a dual supply or a mid-rail reference; single-supply versions need a bias network added to every op-amp's non-inverting input.